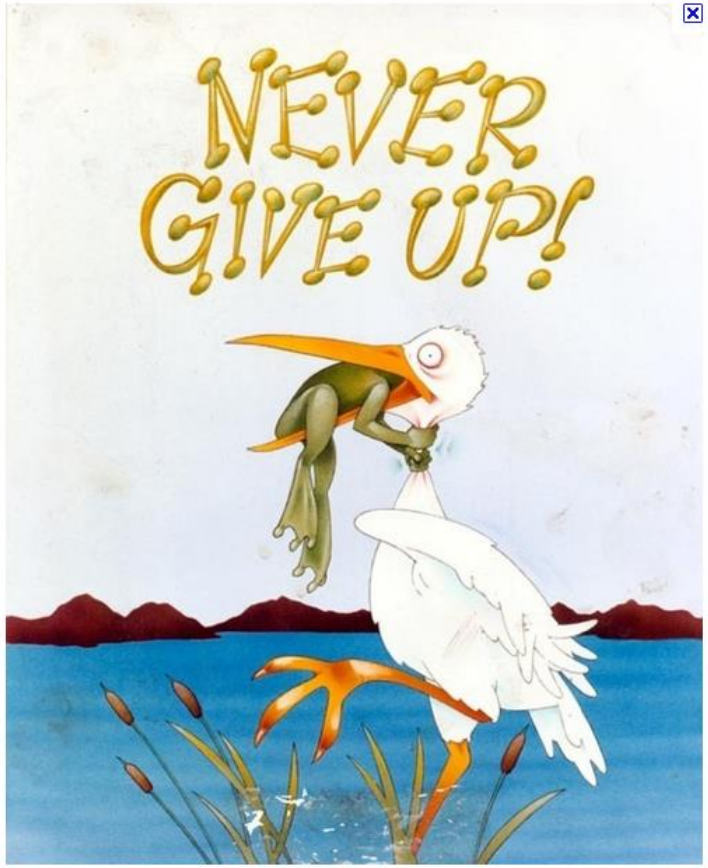
Review Notes

STAT 2793 Winter Term 2026

At times, learning statistics may seem tough.
In those moments remember to...



Big Picture

"STATISTICS" IS ...

STAT 1793

MEAN
MEDIAN
VARIANCE / SD
PERCENTILES

DESCRIPTIVE STATISTICS

BAR PLOT
PIE CHART
HISTOGRAM

GRAPHICAL REPRESENTATION

STAT 2793 (2/3 MDS)

CONFIDENCE INTERVALS
HYPOTHESIS TESTS
↳ ONE MEAN
↳ 2 INDEPENDENT
↳ ANOVA

INFERENTIAL STATISTICS

STAT 2793 (1/3 RD)

REGRESSION
CORRELATION
GENERAL LINEAR MODEL
GENERAL ADDITIVE MODEL

REGRESSION ANALYSIS

STAT 1793

← PROBABILITY →

Terminology (we've hopefully used before)

- The *mean* of a set of data is the sum of the observations divided by the number of observations

$$\bar{x} = \frac{\sum x_i}{n}$$

"OBSERVATIONS" (DATA)

SAMPLE SIZE

- The *median* of a set of data is the observation in the middle of an ordered list $x_1, x_2, \dots x_n$ [ORDERED] (NUMBERS)

M: IF n IS ODD M = MIDDLE ORDERED POINT IF $n=9$; x_5 MEDIAN

IF n IS EVEN M: AVERAGE OF 2 MIDDLE VALUES IF $n=10$

$$M = \frac{x_5 + x_6}{2}$$

Terminology (we've hopefully used before)

- The *variance* and *standard deviation* are measures of the variation in the data. The standard deviation is the square root of the variance.

VARIANCE s^2 (σ^2) "ADD UP"

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1} \quad \text{OR} \quad s^2 = \frac{\sum x_i^2 - \frac{(\sum x_i)^2}{n}}{n-1}$$

↳ UNITS ARE SQUARED ; INTUITION : km^2, m^2, kg^2

STANDARD DEVIATION : $s = \sqrt{s^2}$ INTUITION: UNITS ARE SAME AS ORIGINAL DATA

Terminology (we've hopefully used before)

- The *probability* of an event is a number between 0 and 1, inclusive
  - The notation is $P(E)$ $\quad P(\text{HEADS}) = \frac{1}{2}$
- The *complement* of an event in probability is

  $P(\text{NOT } E)$ or $P(\bar{E})$ or $P(\sim E) = 1 - P(E)$
- Two events are *mutually exclusive* if they do not have any outcomes in common.
  - The addition rule for mutually exclusive events is:

$$P(A \text{ OR } B) = P(A) + P(B)$$

IF EVENTS ARE NOT MUTUALLY EXCLUSIVE: $P(A \text{ OR } B) = P(A) + P(B) - P(A \text{ AND } B)$

Terminology (we've hopefully used before)

- A *random variable* is a quantitative variable whose value depends on chance.

R.V. X, $X = \{x_1, x_2, x_3 \ldots\}$ e.g. $X = \{1, 2, 3, 4, 5, 6\}$ ON A DIE

- A *Probability Distribution* is a listing, graph, or formula denoting the probabilities of all possible outcomes of a random variable.

e.g.

x	0	1	4
$P(x)$	0.5	0.3	0.2

$$P(x) = \binom{n}{x} p^x (1-p)^{n-x}, \quad 0 \le x \le n$$

Terminology (we've hopefully used before)

- *Bernoulli Trials* satisfy the following conditions:
  - Each trial has two possible outcomes, denoted generally as **success** and **failure**
  - The trails are independent
  - The probability of success for the trial *p* is the same from trial to trial
- The *Binomial Distribution* is a discrete probability distribution with parameters *n* and *p*, where *n* is the number of *Bernoulli Trials,* and *p* is the probability of success for each trial.

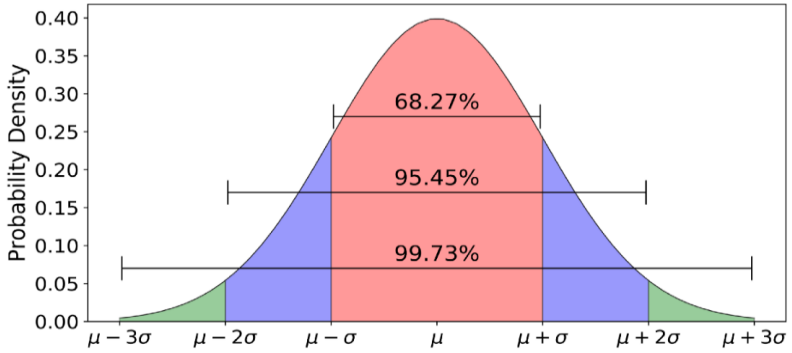
only a set # of outcomes

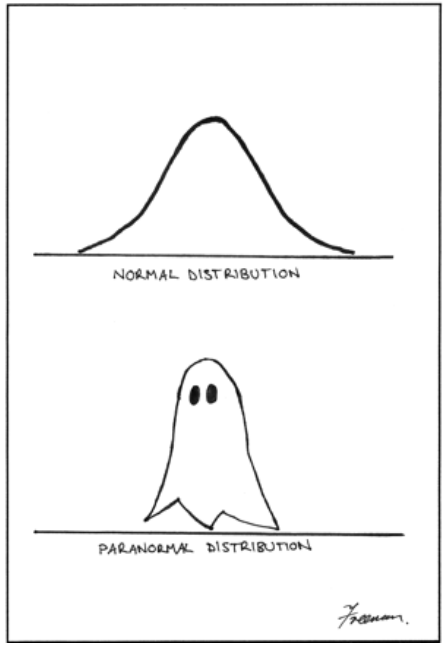
$$P(x) = \binom{n}{x} p^{x} (1-p)^{n-x} \qquad 0 \le x \le n$$

Terminology (we've hopefully used before)

DECIMALS (HEIGHT, DISTANCE, dB)

- The *normal distribution* is a continuous probability distribution with the following characteristics:
  - It is bell-shaped and has a single peak at the center of the distribution
  - The distribution is symmetric
  - It never touches the x-axis
  - It can be described by the mean and standard deviation

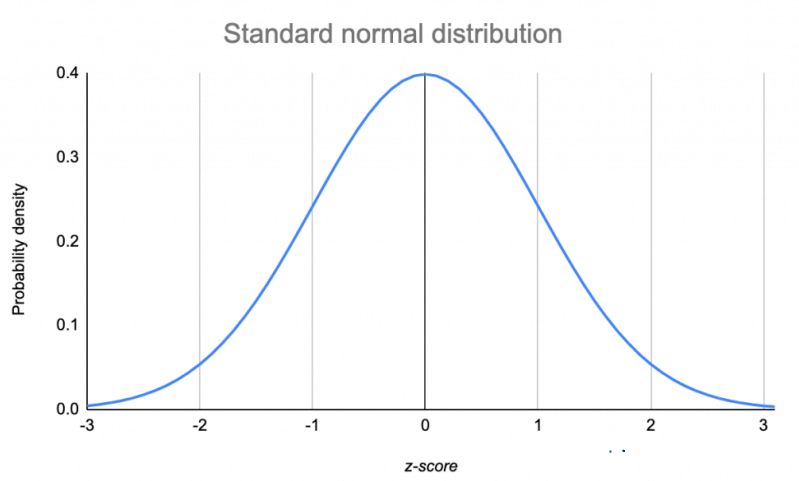


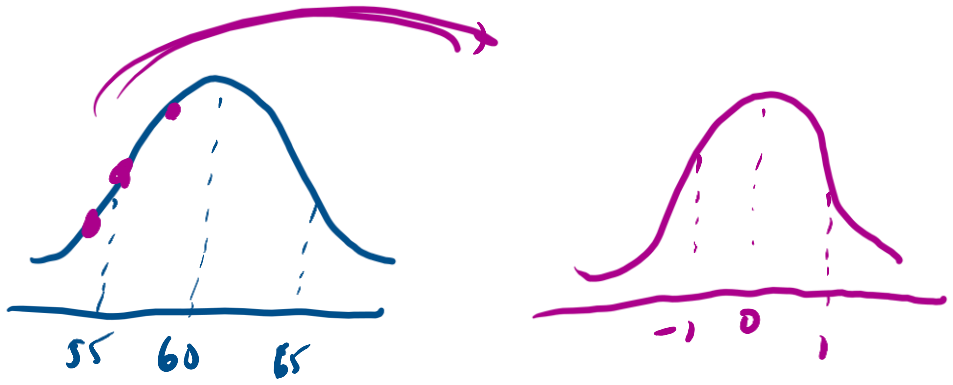


Terminology (we've hopefully used before)

- The *standard normal distribution* is a special type of normal distribution
  - It has a mean of 0 and a standard deviation of 1
  - Any normal distribution can be converted to the standard normal by the following:

$$z = \frac{x-\mu}{\sigma}$$

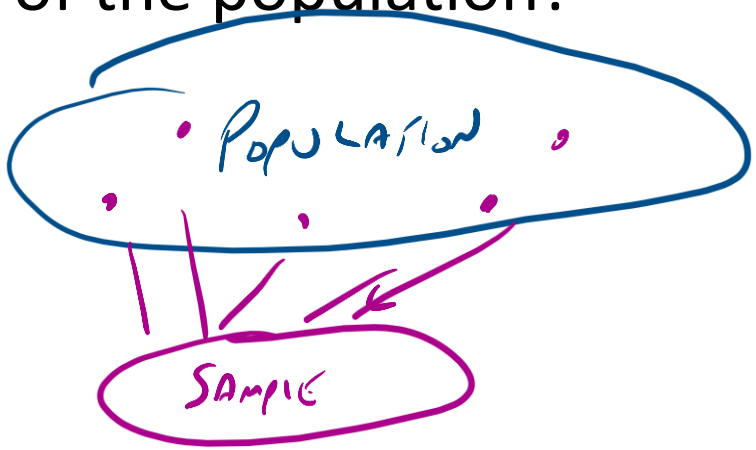




$\mu = 60$, $\sigma = 5$

Terminology (we've hopefully used before)

- A *population* is the set of <u>all</u> measurements or objects of interest in a data set.
- A *sample* is a subset of the population of interest
- If the population is the thing that interests an investigator, why don't they just examine each member/object of the population?
  - Too time-consuming
  - Too costly
  - Impossible in many cases



Terminology (we've hopefully used before)

(GENERALLY NOT KNOWN)

- A *parameter* is a number that describes the population. In statistical practice, the value of the parameter **is not known**, because we cannot examine the entire population

  Examples: μ [MEAN] σ [STANDARD DEV] σ^2 [VARIANCE]

- A *statistic* is a number that can be computed from *sample* data without making use of any unknown parameters. In practice, we often use a statistic to **estimate** an unknown parameter.

  Examples:

  $\bar{x}$ [MEAN] s [STANDARD DEV] s^2 [VARIANCE]

Terminology (we've hopefully used before)

- Samples are used to estimate population characteristics. For example the sample mean ($\bar{x}$) is used to estimate the population mean (μ)

- BUT the sample will not necessarily have a sample mean exactly equal to the population mean. Why? SAMPLE IS ONLY PART OF POPULATION

- This results in *sampling error*: the difference between a sample statistic and its corresponding population parameter.

TRADEOFF → HAVING SAMPLING ERROR & TIME/COST/RESOURCE SAVINGS FROM TAKING THE SAMPLES

- One way to mitigate sampling error is to take large samples and/or MANY samples (sample over and over and over and over and….)

The Sampling Distribution of the Sample Mean ($\bar{x}$) - Facts

- Suppose that $\bar{x}$ is the mean of a simple random sample of size *n* drawn from a large distribution with population mean μ and population standard deviation σ.
  - Then the sampling distribution of the sample mean $\bar{x}$ has mean μ and standard deviation $\frac{\sigma}{\sqrt{n}}$.

- If individual observations are <u>normally distributed</u> with mean μ and population standard deviation σ, then the sample mean
  - Then the sampling distribution of the sample mean $\bar{x}$ is <u>normally distributed</u> with mean μ and standard deviation $\frac{\sigma}{\sqrt{n}}$.

(https://onlinestatbook.com/stat_sim/sampling_dist/)

The Central Limit Theorem

- For a "relatively large" sample size, the variable $\bar{x}$ is approximately normally distributed, regardless of the distribution of the variable under consideration. The approximation becomes better with increasing sample size.
  - Rule of thumb for "relatively large": $n \geq 30$

- The Central Limit Theorem allows us to use normal probability calculations to answer questions about sample means from many observations, even when the *population distribution is not normal.*

(https://onlinestatbook.com/stat_sim/sampling_dist/)

Tools/Techniques we need to remember…

- How to calculate the mean of a sample of values
- How to calculate the standard deviation of a sample of values
- Probability Basics
- Where to find special standard normal table (*z*) values
  - Table II
- The sampling distribution of the sample mean and the impact of those results from https://onlinestatbook.com/stat_sim/sampling_dist/